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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
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Key Personnel Changes/Previous PI:			
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Task Description:

Flame spread over solid fuels in an opposed-flow environment has been investigated for over four decades for understanding the fundamental nature of hazardous fire spread. The appeal for this configuration stems from the fact that flame spread rate remains steady, even if the flame itself may grow in size. For practical fire safety issues, however, wind-assisted flame spread is more relevant.
However, these two regimes have always been studied in isolation without much effort to establish a connection, even though the underlying mechanism of flame spread is the same in all regimes. Sitting between the two regimes are high-residence time flames, as found in a low-velocity or quiescent microgravity environment. Residence time is the time spent by an oxidizer in the combustion zone. Such flames, which are of interest on their own merit due to fire safety issues in spacecrafts, offer some unique characteristics because of the high residence time. Radiation becomes dominant and, based on previous space experiments and analysis, we contend that a vigorously spreading flame on Earth becomes self-extinguishing in a microgravity environment under certain conditions such as the fuel thickness being greater than a critical value.

The proposed research uses a comprehensive approach-- a novel experimental set up and a theoretical framework based on scaling and numerical modeling-- to investigate flame spread driven by varying residence time, from blow-off extinction in an opposed-flow configuration through high residence time flame to blow-off extinction in a concurrent-flow configuration. At the heart of this proposal is a novel but simple experiment where the residence time of the oxidizer can be controlled and high residence time flames can be established for a long duration (compared to drop towers). As a proof of concept, we have constructed a flame tower at San Diego State University (SDSU) in which, after a sample is ignited, the sample holder, placed in an open moveable cart, can be traversed at any desired speed upward or downward, creating an external flow that can augment or mitigate the buoyancy-induced flow. Preliminary results show that we can control the residence time and create flames in different regimes, including a transition between a wind-aided and wind-opposed configuration. At Gifu University in Japan, we have been developing an interferometry based imaging system which we intend to enhance to capture the thermal footprint of a flame's leading edge. The leading edge is central to our understanding of mechanism of flame extinction. Further development of this technology will enable us to integrate diagnostics in future space based experiments and provide validation data to a comprehensive numerical model. The comprehensive model, to be built upon our existing two-dimensional model, will solve an unsteady, three-dimensional, Navier stokes equation with finite rate kinetics in the gas and solid phases and radiation in the gas phase. The software implementation will be object-oriented and utilize a new technology called Web Services that will decouple various sub-models and enhance parallel execution.

The radiation model will also be refined by including the equilibrium composition of species for finding radiative properties in high residence-time flames. The new data generated by our proposed ground-based experiment will be the first of its kind to encompass flame spread across different regimes and will stimulate fundamental theoretical research to unify different regimes of flame spread. The comprehensive model, tested against available theory, data in literature, and data generated at SDSU and Gifu, will be applied to test the three hypotheses presented in this proposal regarding flame extinguishment in a microgravity environment. A successful outcome of this project will lead to a well thought out space-based experiment on the mechanism of flame extinction in a gravity free environment.

NOTE: Project continues as Residence Time Driven Flame Spread Over Solid Fuels--NNX15AG11G.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our research has three components: We have built a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread are being conducted; Theoretical and computational work that explore the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames.
The data that we are acquiring in the Flame Tower will provide the research community with a comprehensive set of results for testing different theories of flame spread in a normal gravity environment. Moreover, by controlling the residence time, various regimes of flame spread, including the microgravity regime, can be explored in the Flame Tower. Our theoretical work predicts a fuel thickness beyond which steady flame spread is unsustainable in a gravity free environment. If we are successful in establishing a critical thickness, this will have a powerful impact on making fire resistant environment for humans in space.

The tasks we have performed to date can be separated into four different categories. Below, we list the progress we are making in each.
A. Ground-Based Experimental Work: The goal of this work is to establish the role of residence time, time spent by an oxidizer in a flame leading edge, on the mechanism and control of flame spread. Towards this goal, we have been building a number of ground-based experiments involving flame spread over thin solid fuels in an opposed flow environment.

A.1: SDSU Flame Tower: The flame tower is the centerpiece of our ground-based activities. We have finished the construction of a 10 m tall steel chamber (details on year-1 report) inside which a fuel sample mounted on a cart can be traversed up or down a vertical rail with a prescribed velocity. We have been successful in developing a completely remote controlled system to move a cart at any desired speed (from -3 m/s to + 3 m/s: details in year-2 report). We have conducted detailed velocity measurement to establish that the flow seen by the flame is uniform upstream over a 40 mm by 40 mm area upstream of the fuel sample (which is 20 mm wide).

The design and operation of the micro flame tracker, which is housed inside the moving cart, has been described in the year-2 report. Once the flame is ignited, a gas phase thermocouple, attached to a linear motion system on the cart, tracks its motion of the leading edge of the flame, providing the instantaneous flame spread rate. The flame spread is also obtained by analyzing the side-view digital video of the flame, allowing us to verify the data from the automated tracking system.

Data on flame spread rate and flame shape were obtained for flame spread over ashless filter paper with the relative flow velocity varying from positive (opposed flow configuration) to negative (concurrent flow configuration). The spread rate behavior was consistent with theoretical prediction for the opposed flow configuration. When the cart was moved upward (in the same direction of the buoyancy driven flow), the flame spread rate remained fairly constant (or slightly

Task Progress:

increasing) until about a flow speed of -40 cm/s, when the flame converted itself into its concurrent-flow configuration (wind assisted flame spread). We are still analyzing the wealth of data produced by these experiments.

The flow velocity at which blow-off extinction occurs was found to be sensitive to the development length of the boundary layer. Using an air flow sensor, we characterized the velocity field seen by our moving sample. A detailed study using Fluent was used to relate that cart velocity with the velocity seen by the flame (see year-4 report).

The effect of the flow velocity and the boundary development lengths were experimentally studied using ashless filter paper and the results strongly support an effective velocity correlation that we developed from scale analysis (see details in year-4 report).

A.2: Flame Stabilizer: One of the challenges in the experimental study of flame spread is that, even if the flame spreads at a steady rate, the propagating flame creates an unsteady phenomenon with respect to the laboratory frame of reference. As a result, it is difficult to obtain detailed data, necessary for validating models, in a spreading flame. To remedy this situation, we have built a novel flame spread apparatus that moves the fuel in the opposite direction of the flame spread to keep the leading edge of the flame stationary with respect to the laboratory. A thermocouple, fixed to the laboratory frame of reference, in front of the leading edge of the flame senses the presence of the flame and a PID controller keeps its temperature constant by moving the sample holder, driven by a stepper motor, in the opposite direction at the velocity of the spread. Instantaneous flame spread rate and the visible flame structure are compared for a downward spreading flame over ashless filter paper with the corresponding stationary flame. The results indicate that the difference between the two configurations are within experimental uncertainties and the stabilized flame can represent a spreading flame adequately, including variability of flame spread rate and the flame geometry, for further observation. We have presented this work in the 34th International Symposium on Combustion. With a spreading flame stabilized by this apparatus, we are in a position to measure gas phase temperature, including in the plume region, where fluctuations due to turbulence makes it very difficult to map out the thermal field of a spreading flame.

Using a K type thermocouple we mapped the gas phase temperature field of a stabilized flame. An infrared CO2 sensor was used to map out the CO2 concentrations.

When the temperature and CO2 concentrations are normalized by their equilibrium values (0 for ambient conditions and 1 for chemical equilibrium values), the similarity between the temperature and CO2 is remarkable. Using the fluctuations in the signal, the pseudo-turbulence intensity was calculated for both the temperature and CO2 concentrations showing strong similarity. Turbulence is most intense at the far downstream of the flame and in the outer zone of entrainment.

B. Preparation for Space-Based Experimental Work: We have refined the objectives of our space-based experiments and produced a concept design for the proposed experiment. The objectives for the space-based experiments are the following: a. Establish that steady flame spread over solid fuel in a quiescent microgravity environment is possible only if the fuel thickness is less than a critical value. b. Monitor the propagation of the products and temperature field to test the hypothesis that radiative extinction occurs when a flame is vitiated by its own product fields. c. Determine how the flame spread rate and the extinction behavior depend on the ambient pressure. d. Determine how the flame spread rate and the extinction behavior depend on the developing boundary layer when a mild opposing flow is present.

The basic concept of the proposed experiment is as follows: A carousel driven by a stepper motor holds several sample holders, each of which are placed in the test zone by rotating the carousel. Samples of different thicknesses are held in each sample holder, which carries its own igniter. When a sample is rotated into the test zone, the igniter snaps into the ignition circuit. As the flame spreads over the sample, gas phase thermocouples and CO2 sensors monitor the propagation of the thermal diffusion fronts. Once the propagating flame reaches the end of the sample or extinguishes, the recording of the temperature and species field will continue to provide valuable data on transient cooling of the gas phase and the diffusion of species. After a preset interval, the next sample is rotated into position and the experiment is repeated for a new fuel thickness. The carousel will then be replaced with a second carousel and the ambient pressure will be reduced, which is predicted to be a significant parameter just like the fuel thickness in determining the critical fuel thickness.

We have proposed an experimental matrix in the BASS-II project to that will help us determine (a) a suitable ignition method; (b) select an oxygen level suitable for flame spread over poly(methyl methacrylate) - PMMA ; (c) estimate extinction time at lower oxygen level and; and (d) evaluate the width effect to supplement our original experimental matrix in the SoFIE project.

C. Theoretical/Modeling Work: We are continuing to make progress in modeling flame spread over solid fuels under different conditions. Our modeling/theoretical effort can be summarized as follows:

1. We have been developing Web based tools for calculating equilibrium temperature of PMMA and Cellulose combustion. This calculation tools, which can be used by the community, helps us determine exactly how much sample burn is possible (under different conditions) in a closed chamber without significantly altering the oxygen level. It also predicts the equilibrium composition providing us with the thermodynamic limits of CO2 level and temperature in the gas phase to be expected.

2. We are using a two dimensional model with finite-rate one-step kinetics, and radiation to simulate opposed flow flame spread. The model has been used to compare downward flame spread results with experiments conducted in the lab. The spread rate from the model for three different fuel thicknesses agreed quite well with the experimental results for downward spread over PMMA sheets under the ambient conditions.

3. The model was used to compare pure downward flame spread with the stabilized flame produced by our stabilizer device. The comparison of the numerical results as well as experimental data established the flame stabilizer does reproduce all the characteristics of a downward spreading flame, only the flame is now frozen in the laboratory coordinate ready for prolonged examination. This study established the flame stabilizer as a new viable platform for experimental studies of flame spread.

4. The data from the flame tower showed that the blow off velocity (of the opposing flow) was related to the boundary layer development length. The computational model, along with scale analysis, was used to quantify the effect of the development length in terms of an effective velocity. An effective velocity for a flame, embedded in a boundary layer, is defined as an equivalent velocity seen by the flame. The effective velocity is then correlated with free stream velocity, development length of the boundary layer, and fluid and fuel properties. The resulting correlations were remarkably

	accurate in explaining the blow off extinction velocity over a wide range of parametric conditions. 5. We have done detailed radiation calculations to establish the importance of radiation loss versus radiation feedback. Also, the radiation loss correction of thermocouple measurement has been computed taking into account both radiative loss and gain by the thermocouple bead. D. Space Based Experiments (BASS-II): We have conducted three sets of experiments as part of the BASS-II project, burning twenty samples of PMMA. Results from these experiments are still being analyzed, producing two conference articles and several journal articles, which are in preparation. E. Dissemination of Results: We have published 8 journal papers, one textbook, and many conference papers. A research website, http://flame.sdsu.edu , has been created to highlight our accomplishments and share our research results with our peers. NOTE: Project continues as Residence Time Driven Flame Spread Over Solid Fuels--NNX15AG11G.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Bhattacharjee S, Carmignani L, Bussett K, Celniker G. "The Effect of Boundary Layer on Blow - Off Extinction: Experiments and Simplified Analysis." 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Bhattacharjee S, Nadertaber A, McGrath K, Ivisic I. "Steady Spread and Radiative Extinction in Opposed - Flow Flame Spread over Thin PMMA in a Microgravity Environment: Results from BASS - II Experiments." 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Ferkul P, Olson S, Bhattacharjee S, Fernandez-Pello C, Miller F, Tien J. "Combustion of solids in Microgravity : Results from the BASS - II Experiments." 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 7th International Symposium for Physical Sciences in Space and 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Bhattacharjee S, Tran W, Laue M, Paolini C, Nakamura Y. "Experimental validation of a correlation capturing the boundary layer effect on spread rate in the kinetic regime of opposed-flow flame spread." Proceedings of the Combustion Institute. 2015;35(3):2631-8. http://dx.doi.org/10.1016/j.proci.2014.06.125 , Jan-2015
Articles in Peer-reviewed Journals	Bhattacharjee S, Paolini C, Tran W, Villaraza JR, Takahashi S. "Temperature and CO2 fields of a downward spreading flame over thin cellulose: A comparison of experimental and computational results." Proceedings of the Combustion Institute. 2015;35(3):2665-72. http://dx.doi.org/10.1016/j.proci.2014.05.093 , Jan-2015
Articles in Peer-reviewed Journals	Takahashi S, Ebisawa T, Bhattacharjee S, Ihara T. "Simplified model for predicting difference between flammability limits of a thin material in normal gravity and microgravity environments." Proceedings of the Combustion Institute. 2015;35(3):2535-43. http://dx.doi.org/10.1016/j.proci.2014.07.017 , Jan-2015
Books/Book Chapters	Bhattacharjee S. "Thermodynamics: An Interactive Approach." New York: Pearson Education, Inc., 2014., Nov-2014
Papers from Meeting Proceedings	Carmignani L, Celniker G, Busset K, Paolini C, Bhattacharjee S. "Two different approaches for creating a prescribed opposed-flow velocity field for flame spread experiments." EFM14 – Experimental Fluid Mechanics 2014, Czech Republic, Nov 18-21 , 2014. EPJ Web of Conferences. 2015;92:02011. http://dx.doi.org/10.1051/epjconf/20159202011 , May-2015